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Analytical Biochemistry 365 (2007) 246–259
ANALYTICAL BIOCHEMISTRY
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# Monitoring repair of DNA damage in cell lines and human peripheral blood mononuclear cells

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Received 20 November 2006
Available online 21 March 2007

## Abstract

We introduce a method to follow DNA repair that is suitable for both clinical and laboratory samples. An episomal construct with a unique 8-oxoguanine (8-oxoG) base at a defined position was prepared in vitro using single-stranded phage harboring a 678-bp tract from exons 5 to 9 of the human P53 gene. Mixing curve experiments showed that the real-time PCR method has a linear response to damage, suggesting that it is useful for DNA repair studies. The episomal construct with a unique 8-oxoG base was introduced into AD293 cells or human peripheral blood mononuclear cells, and plasmids were recovered as a function of time. The quantitative real-time PCR assay demonstrated that repair of the 8-oxoG was 80% complete in less than 48 h in AD293 cells. Transfection of small interfering RNAs down-regulated OGG1 expression in AD293 cells and reduced the repair of 8-oxoG to 30%. Transfection of the episome into unstimulated white blood cells showed that 8-oxoG repair had a half-life of 2 to 5 h. This method is a rapid, reproducible, and robust way to monitor repair of specific adducts in virtually any cell type.


*Keywords:* DNA damage; Oxidative damage; 8-Oxoguanine; Strand-specific repair; siRNA

Exogenous and endogenous agents constantly inflict DNA damage [1–7], and reactive oxygen species (ROS)[1] are a major source of endogenous damage [6,8]. ROS form a number of damaged bases, but one of the most diagnostic bases for this type of damage is 8-oxoguanine (8-oxoG) [9,10]. When 8-oxoG is in DNA, it miscodes for adenine, producing G $\rightarrow$ T transversion mutations when DNA synthesis occurs [11–13], and mice lacking 8-oxoG and MYH develop tumors [14–20]. In addition, accumulation of 8-oxoG has been associated with lung tumors that could arise due to DNA repair defects [21,22].

To eliminate 8-oxoG and its significant consequences, cells have evolved repair systems to eliminate damage and prevent genotoxicity [23,24]. Base excision repair (BER) is a major factor in the repair of 8-oxoG [25–31]. In human cell extracts in vitro, OGG1 is the major DNA glycosylase that excises 8-oxoG and initiates the BER pathway [29,32,33], but 8-oxoG can also be repaired to a lesser extent by a series of DNA glycosylases known as NEIL1, NEIL2, and NEIL3 [29,34–36]. Despite the fact that

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[1] *Abbreviations used:* ROS, reactive oxygen species; 8-oxoG, 8-oxoguanine; BER, base excision repair; NER, nucleotide excision repair; siRNA, small interfering RNA; ODN, oligodeoxyribonucleotide; MALDI, matrix-assisted laser desorption/ionization; cDNA, complementary DNA; RF, replicative form; ATCC, American Type Culture Collection; DMEM, Dulbecco's modified Eagle's medium; ccc, covalently closed circular; RT–PCR, reverse transcriptase PCR; PBMC, peripheral blood mononuclear cell; PBS, phosphate-buffered saline; GFP, green fluorescent protein; BSA, bovine serum albumin; DMSO, dimethyl sulfoxide; PVDF, polyvinylidene fluoride; IgG, immunoglobulin G; dsDNA, double-stranded DNA; ssDNA, single-stranded DNA; AP, apurinic/apyrimidinic; mRNA, messenger RNA.

0003-2697/$ - see front matter 
doi:10.1016/j.ab.2007.03.016

OGG1 plays a major role in excising 8-oxoG, OGG1 excises 8-oxoG only from the nontranscribed strand of transcribed genes [37,38]. In vitro nucleotide excision repair (NER) assays show that 8-oxoG is subject to excision, consistent with recognition by that repair system [39]. There is also evidence that mismatch repair could be involved in the removal of 8-oxoG adducts [40,41]. Therefore, a number of DNA repair pathways could have a role in the eradication of 8-oxoG to limit possible genotoxic effects.

One obstacle to the study of DNA repair of oxidative damage in mammalian systems is that it is pleiotropic, reacting not only with DNA but also with proteins and membranes [20,42,43]. Despite the fact that the number of adducts produced by oxidative damage is large, progress has been made toward elucidating how repair occurs in cells. Repair of oxidative damage can be monitored using numerous techniques, including mass spectrometry, PCR, and alkaline elution [44–47]. All of these techniques monitor DNA damage in cells, but the wide range of targets for oxidative damage can obscure the role of DNA repair, leading to mutation. Targeted deletion or small interfering RNA (siRNA) depletion, which either obliterates or reduces OGG1 expression in mouse or human cells, has shed light on the role of OGG1 in preventing mutation [16,19,48–53]. Another useful assay, based on OGG1 activity in cell extracts, has also provided interesting data [54,55].

One way to monitor damage directly in cells is through host cell reactivation assays that have been used to follow repair of several types of DNA damage in stimulated lymphocytes or lymphoblasts [56–66]. Differences in normal and NER-deficient cells have been detected using this method. These assays are useful, but generally the plasmids are modified in vitro with DNA damaging agents; therefore, repair is observed only indirectly by monitoring an enzymatic activity such as chloramphenicol acetyltransferase or luciferase. Nonetheless, these techniques can be used to investigate larger populations, although they are limited in providing details about DNA repair at the molecular level and the adducts removed from the DNA because the assay is based on the protein produced and so requires the disruption of the transcribed coding sequence or promoter.

Instead of subjecting cells to oxidative damage at a global level or by the introduction of globally damaged plasmid by transfection, another way to study DNA repair is through the use of episomal vectors that can be isolated from mammalian cells and then probed for repair. We can insert specific modified bases at defined sites and monitor DNA repair or mutagenesis [67–73]. To determine repair kinetics, a nonreplicating episome is transfected into mammalian cells, and then the closed circular episomal DNA is harvested. One method to evaluate repair of site-specific adducts uses Southern blotting [37,74]. There are definite advantages to using Southern blotting to monitor DNA repair at a specific site, but it is still more difficult to examine replicate samples. Therefore, the capacity to monitor repair of a unique adduct in episomal DNA reproducibly would allow researchers to follow the kinetics for the elimination of damage in small populations (10–100 persons). This would yield information on the specific defects in DNA repair systems.

In this article, we demonstrate a simple method to follow DNA repair of episomal DNA in any type of cells. This real-time PCR-based method is rapid and reproducible compared with existing methods. It should have applications in the field of DNA repair and be useful in the determination of defects in human cells. To show its utility, we measured repair of 8-oxoG adducts in both cell lines and normal unstimulated human lymphocytes. Therefore, this assay could be used to monitor DNA repair in patient samples. We also used siRNA to knock down the production of OGG1 in human cells, and we show that this method is suitable for examination of repair of 8-oxoG in nontranscribed DNA.

## Materials and methods

### *Chemicals and enzymes*

Oligodeoxyribonucleotides (ODNs) were obtained from IDT (Coralville, IA, USA) or Midland Certified Reagent (Midland, TX, USA). All modified 8-oxoG ODNs were verified using matrix-assisted laser desorption/ionization (MALDI). siRNAs were prepared using the Silencer predesigned siRNA kit (Ambion, Austin, TX, USA). All ODNs and siRNAs used in this study are indicated in Table 1. Fpg was obtained from laboratory stocks and used as described [75–77]. RB69 DNA polymerase expression vector was a kind gift from William Konigsberg (Yale University), and the DNA polymerase was purified using established protocols [78–80]. T4 DNA ligase was obtained from New England Biolabs (Beverly, MA, USA) or was cloned and purified in the laboratory as described below. Pfu DNA polymerase was purified from an overexpressing clone prepared by Bin Wang in the laboratory using established methods. T4 polynucleotide kinase was obtained from Roche (Indianapolis, IN, USA). $\gamma$-[$^{32}$P]ATP was obtained from ICN Biochemicals (Costa Mesa, CA, USA). The OGG1 antibody was obtained from Pablo Radicella and Serge Boiteux (CNRS–CEA, Fontenay-aux Roses, France), and the tubulin antibody was obtained from Santa Cruz Biotechnology (Santa Cruz, CA, USA). The M13mp18-P53 was prepared by insertion of a 678-bp PCR fragment from the P53 complementary DNA (cDNA) from 481 to 1159 into the *Bam*HI/*Xba*I site in replicative form (RF) M13mp18. That region spans exons 5 to 9 of the P53 cDNA. The identity of the insert was verified by DNA sequencing. Bovine pancreatic DNase I grade II (Roche) and RNase A (Sigma, St. Louis, MO, USA) were used as suggested by the respective manufacturers. Lysozyme was obtained from Sigma. All molecular biological or biochemical methods not explicitly described were performed using standard protocols [81].

Table 1
ODNs used in this study

| ODN | Sequence[a] | Use[b] |
|---|---|---|
| CMH3 | AGCTCTCGGATCCTCTCGAA | Extension assay |
| CMH5 | ATGAGCGCTGCTCAGATAG | Assay |
| CMH6 | GGTGGTACAGTCAGAGCCAAC | Linear/Assay |
| CMH7 | GTGACGAAACTCAGTGTTAC | Assay |
| CMH8 | TCGAGAGGGTTGATATAAG | Linear/Assay |
| LIGFor | GCCGCCATGGTTCTTAAAATTCTGAAC | *Nco*I |
| LIGRev | GCGCGGATCCCATAGACCAGTTACCTCA | *Bam*HI |
| OGG1 siRNA 214 | GAGGUGGCUCAGAAAUUCCtt | Sense |
| | GGAAUUUCUGAGCCACCUCtt | Antisense |
| OGG1 siRNA 330 | GCUUGAUGAUGUCACCUACtt | Sense |
| | GUAGGUGACAUCAUCAAGCtg | Antisense |
| OGG1 siRNA 366 | GCUAGAUGUUACCCUGGCUtt | Sense |
| | AGCCAGGGUAACAUCUAGCtg | Antisense |
| Nonspecific SiRNA control | ttAUGUUGACGUGUUCCAAAU | Sense |
| | ttAUUUGGAACACGUCAACAU | Antisense |
| 8-oxoG | PTGGTGGTGCCCTAT$^{\mathbf{8}}$**G**AGCCGCCTGA | DNA synthesis |
| OGG1 RT–PCR | CGTGGACTCCCACTTCCAAGAG | Forward |
| | ATGGTAGGTGACATCATCAAGCT | Reverse |
| GAPDH RT–PCR control | CCAAGGTCATCCATGACAAC | Forward |
| | CACCCTGTTGCTGTAGCCA | Reverse |

*Note.* All ODNs shown are $5' \rightarrow 3'$ in polarity.

[a] The underlined sequences represent two mutations found in the sequenced exons 5 to 9 for the ODN that was used for primer extension analysis located at position 1147 to 1166 in the NCBI database—AF307851 (nucleotide).

[b] Extension assay: primer extension to show position of 8-oxoG; assay: use for exponential amplification; linear: primer for extension in the first step of the assay; *Nco*I: inserts an *Nco*I site on the 3′ end of the T4 DNA ligase gene; *Bam*HI: inserts a *Bam*HI site on the 3′ end of the T4 DNA ligase gene; sense: mRNA sequence; antisense: siRNA for the antisense strand; DNA synthesis: to insert a modified base; forward: primer for RT–PCR; reverse: primer for RT–PCR.

*Cell culture*

The mammalian cell line AD293, a transformed human kidney cell line, was obtained from American Type Culture Collection (ATCC, Manassas, VA, USA) and was grown in Dulbecco's modified Eagle's medium (DMEM) supplemented with 10% fetal bovine serum at 37 °C under a 5% $CO_2$ atmosphere. Cells were regularly passaged to maintain exponential growth. Synthetic siRNA was transfected using Lipofectamine 2000 (Invitrogen, Carlsbad, CA, USA) according to the manufacturer's protocol.

*Isolation of T4 DNA ligase coding sequence and purification of T4 DNA ligase*

*Escherichia coli* phage T4 was obtained from ATCC, and phage DNA was purified by phenol/$CHCl_3$ extraction. A PCR fragment of full-length cDNA of T4 DNA ligase was prepared with Pfu DNA polymerase and primers (Table 1). The restriction endonuclease cleaved PCR products were subcloned into the *Nco*I and *Bam*HI sites of a pET-28a(+)-modified plasmid with a tobacco etch virus protease site in the synthesized protein with a His6 tag at the C-terminal end (Novagen, Madison, WI, USA, and Yuan Chen, City of Hope National Medical Center, Duarte, CA, USA). Standard expression and purification methods were used to prepare the T4 DNA ligase. Protein purity was monitored by SDS–PAGE. The activity of T4 DNA ligase was compared with that of commercial preparations using an assay ligating *Hin*dIII fragments of γ DNA at different dilutions and separating products on agarose gels.

*In vitro synthesis of RF M13mp18-P53-8-oxoG*

A double-stranded, closed circular molecule with a unique 8-oxoG adduct was formed using in vitro DNA synthesis and ligation. Briefly, for DNA synthesis, 15 pmol (130 ng) of the kinased 8-oxoG ODN (Table 1) was annealed to the single-stranded M13mp18-P53 (7 μg) by heating to 80 °C and cooling to 30 °C slowly over 30 min. Synthesis and ligation were conducted on the annealed product using the RB69 DNA polymerase (1.4 μg) and T4 DNA ligase (400 U) in the presence of 2.5 mM ATP and 150 μM of each dNTP. The reaction mixture was incubated using the following temperatures and times: 4 °C for 5 min, 22 °C for 5 min, and 30 °C for 3 h. The closed circular plasmid was purified using a CsCl gradient (1 g/ml) in 0.4 mg/ml ethidium bromide followed by extraction with water-saturated butanol, dialyzed against 10 mM Tris–HCl (pH 7.5) and 1 mM EDTA for 6 h, ethanol precipitated for 16 h at 4 °C, centrifuged, washed with 70% ethanol, and resuspended in 10 mM Tris–HCl (pH 7.5) and 0.1 mM EDTA at a concentration of 1 mg/ml. To synthesize sufficient RF DNA, either a number of reactions or large-scale reactions were performed, and the DNA was pooled prior to CsCl gradient purification of supercoiled DNA.

*Characterization of 8-oxoG ccc DNA*

After making 8-oxoG RF M13mp18-P53-8-oxoG, we confirmed that more than 90% of the DNA was covalently closed circular (ccc) with a single 8-oxoG in the region for analysis using primer extension. After treatment with Fpg, the RF M13mp18-P53-8-oxoG DNA was purified by phenol/$CHCl_3$ extraction and ethanol precipitation. The extension reaction used 1× Pfu buffer (Stratagene, La Jolla, CA, USA), 200 μM of each dNTP, 20 μM single primer CHM3 (5′-$^{32}$P-end-labeled using T4 polynucleotide kinase) (Table 1), and Pfu polymerase in 100 μl total reaction volume at 94 °C for 3 min, followed by 30 cycles of 94 °C for 40 s, 53 °C for 40 s, 72 °C for 1 min, and 72 °C for 10 min extension. Products of the primer extension reaction were separated on a DNA sequencing gel and imaged using a Typhoon 9410 PhosphorImager system (GE Health Sciences, Piscataway, NJ, USA).

*siRNA knockdown of OGG1 expression in AD293 cells*

Briefly, 24 h before transfection, AD293 cells were plated on 100-cm plates at 80 to 90% confluence. For transfection, 50 μl Lipofectamine 2000 diluted in 1.5 ml OptiMEM (Invitrogen) was incubated for 5 min at room temperature. At 5 min, per plate, 520 pmol double-stranded siRNA in 1.5 ml OptiMEM was mixed with the Lipofectamine 2000–OptiMEM mixture for each plate and then applied to the cells. At 6 h posttransfection, each plate was supplemented with 7 ml DMEM with 10% fetal calf serum. Initial experiments verified that the knockdown 48 h posttransfection was nearly 100%. At 48 h, another transfection of AD293 cells was conducted, and the knockdown was verified at 96 h following the first transfection. The knockdown of OGG1 production was verified using Western blotting of OGG1. As a control, OGG1 expression in AD293 cells was compared with that of GAPDH and OGG1 production was compared with that of tubulin.

*RT–PCR*

Total RNA was extracted using TRIzol reagent (Invitrogen) according to the manufacturer's protocol. Gene expression was analyzed using reverse transcriptase PCR (RT–PCR). First-strand cDNA was synthesized from 2 μg total RNA using SuperScript reverse transcriptase II (Invitrogen). For OGG1 expression, primers were added (Table 1), and the amplification was carried out at 94 °C for 40 s, 54 °C for 40 s, and 72 °C for 1 min for 26 cycles [81]. For analyzing GAPDH gene expression, primers were added (Table 1), and the reactions were performed at 94 °C for 40 s, 53 °C for 40 s, and 72 °C for 1 min for 23 cycles.

*In vivo assay for 8-oxoG repair*

*Liposome transfection in AD293 cells*

Here 4 μg ccc RF M13mp18-P53-8-oxoG was transfected into $1 \times 10^6$ AD293 cells on a 100-mm plate (60% confluence) using Lipofectamine 2000. Cells were then incubated for 3 to 48 h and harvested.

*Nucleofection of PBMCs*

Peripheral blood mononuclear cells (PBMCs) were isolated by Ficoll–Hypaque (Sigma) density gradient centrifugation (specific gravity 1.077) for 30 min at 400*g*. Nucleofection was performed using a Nucleofector device (Amaxa, Gaithersburg, MD, USA) and the human T-cell Nucleofector kit (cat. no. VPA-1002) following procedures recommended by the manufacturer. Cells were washed with phosphate-buffered saline (PBS), and $5 \times 10^6$ PBMCs per nucleofection reaction were transferred to microcentrifuge tubes, centrifuged, and resuspended in 100 μl of the Nucleofector solution and 2 μg of the plasmid DNA being tested. Approximately 1 ml whole blood produced $1 \times 10^6$ PBMCs; therefore, 5 ml whole blood produced the quantity of cells needed for transfection. Cells were then transferred to a nucleofection cuvette, which was inserted into the Nucleofector and pulsed using the Nucleofector program (program U-14). Cells were then immediately transferred to prewarmed 500 μl RPMI and kept at 37 °C for 15 min. Subsequently, cells were transferred to culture dishes containing prewarmed Iscove's modified Dulbecco's medium with 20% fetal bovine serum and incubated at 37 °C for 2, 4, 6, 16, and 24 h postnucleofection as discussed later in Results. Studies using the pmaxGFP plasmid expressing green fluorescent protein (GFP) demonstrated that this procedure resulted in transfection of $22.0 \pm 4.3\%$ of cells based on flow cytometry analysis performed 24 h after nucleofection.

*DNA repair assay using siRNA knockdown of OGG1*

For siRNA knockdown experiments of OGG1 and the plasmid assay, a transfection of the OGG1–siRNA 214 (Table 1) was performed as described above. The only difference is that at 2 days posttransfection, the second transfection of OGG1–siRNA 214 was performed, but 4μg RF M13mp18-p53-8-oxoG was added to the OptiMEM along with the OGG1–siRNA 214 for transfection. Cells were incubated as a function of time and harvested, and DNA was extracted to monitor in vivo repair as for the cells without siRNA treatment.

*DNA isolation*

Any contaminating extracellular DNA was eliminated by treatment of cell cultures with DNase I (40 μl PBS) before DNA isolation, and following thorough washing of the cells, nuclei were isolated [82]. Briefly, $1 \times 10^6$ cells were lysed by the addition of 2 ml of the following buffer: 0.3 M sucrose, 60 mM KCl, 15 mM NaCl, 60 mM Tris–HCl (pH 8.0), 0.5 mM spermidine, 0.15 mM spermine, 2 mM EDTA, and 0.5% Ipegal. The cells with the buffer were incubated on ice for 5 min, followed by centrifugation for 5 min at 1000*g* and 4 °C to pellet the nuclei. Nuclei were washed once with the buffer without Ipegal, and extrachromosomal RF M13mp18-P53-8-oxoG DNA was recovered from nuclei by a small-scale alkaline lysis method [83].

*Fpg treatment of episomal DNA*

Recovered RF M13mp18-P53-8-oxoG DNA (25 ng) was treated with *E. coli* Fpg protein (400 ng) in a buffer of 100 mM KCl, 70 mM Hepes–KOH (pH 7.5), and 0.5 mM EDTA for 1 h at 37 °C to remove any remaining 8-oxoG and introduce a break in the unrepaired closed circular DNA. These conditions result in cleavage of all the 8-oxoG in the RF DNA. Complete cleavage of the 8-oxoG-containing DNA is necessary for the assay success; otherwise, residual 8-oxoG could allow the formation of some products that would falsely indicate repair. The technique used for DNA isolation has been shown to allow the Fpg to function without inhibition, and the quantity of Fpg yields 100% cleavage based on isolation of plasmids with known amounts of 8-oxoG.

*Determination of episomal DNA concentration*

We determined the concentrations of RF M13mp18-P53-8-oxoG isolated using a standard sample concentration program in the DNA Engine Opticon (MJ Research, Waltham, MA, USA). An unknown DNA sample from the isolation of the extrachromosomal DNA was diluted from 1- to 10-fold for template, and standard samples were 2, 1, 0.5, 0.1, 0.05, and 0.01 µg. Standard PCR was carried out in 25-µl reactions and analyzed in triplicate. Briefly, it used 10× Pfu buffer (200 mM Tris–HCl [pH 8.3], 20 mM $MgSO_4$, 10 mM KCl, 100 mM $(NH4)_2SO_4$, 1% Triton X-100), 200 µg/ml bovine serum albumin (BSA, New England Biolabs), 10% dimethyl sulfoxide (DMSO), 200 µM of each dNTP, 400 µM internal control primer set (CMH7 and CMH8, Table 1), 1 U Pfu polymerase, 1.3 µl SYBR Green (diluted 1:1000 in TE buffer [pH 8.0], Roche) and diluted template. Except for the different templates, mixtures of the remaining agents were made and distributed to Eppendorf tubes on ice and then were added to the template samples. PCR mixtures were distributed to hard shell, thin-walled skirted microplates (MJ Research) at room temperature and covered with Microseal "B" Adhesive Film (MJ Research). Conditions of standard PCR were as follows: 94 °C for 3 min, followed by 40 cycles of 94 °C for 40 s, 53 °C for 40 s, 72 °C for 1 min, and 82 °C for 2 s for plate reading (for fluorescence detection). After completion of the PCR, DNA Engine Opticon data were used to determine the concentration of the unknown samples.

*Linear PCR amplification*

The first step was a linear PCR amplification where DNA synthesis was arrested if there was a DNA break. The extended DNA was then subject to real-time PCR amplification. Once the amount of DNA was determined using the standard curve, linear extension was performed on 20 pg template DNA with 20 pmol (175 ng) CMH6 ODN (Table 1), which uses the strand containing the adduct or the break as the template in 50 µl total volume. Conditions for linear PCR were 94 °C for 3 min, followed by 15 cycles of 94 °C for 40 s, 53 °C for 40 s, 72 °C for 1 min, followed by incubation at 72 °C for 10 min. A linear amplification of the M13mp18 internal control region was performed using the CMH8 ODN with conditions identical to those for the CMH6 linear amplification reaction.

*Exponential real-time PCR amplification*

To determine the repair rate in cells, exponential real-time PCR was carried out using aliquots from the primer extension reactions. From the linear amplification reactions, 1 µl was removed and 20 pmol CMH5 and CHM6 primers (Table 1) was added (full-length product 158 bp). Other reagents used were the same as those for standard PCR in a total volume of 25 µl. The reaction conditions were 94 °C for 3 min, followed by 40 cycles of 94 °C for 40 s, 53 °C for 40 s, and 72 °C for 1 min, with the plates read using fluorescence detection. A parallel control real-time PCR reaction for the control DNA region using the CMH7 and CMH8 primers (full-length product 178 bp, Table 1) with 1 µl of the linear amplification reaction was also performed. Real-time PCR was carried out using a DNA Engine Opticon (MJ Research). After PCR, data analysis was performed using Microsoft Excel. Repair rates were analyzed using a ratio of damage region/control region to calculate percentage repair.

*In vitro mixing experiment*

A mixing curve was used to demonstrate the assay linearity. Using RF M13mp18-p53 DNA (100% repair, closed circular) and nicked RF M13mp18-P53-8-oxoG DNA (100% nicked by Fpg), the RF M13mp18-P53-9:nicked DNA ratio was varied from 0 to 1 using a total of 400 pg DNA for each amplification. The mixtures were subjected to analysis using the DNA repair assay described above.

*Western blotting*

Cells were washed with PBS and collected by scraping. Lysis was performed in ice-cold homogenization buffer (1× PBS, 1% Ipegal CA-630, 0.5% sodium deoxycholate, and 0.1% SDS). After incubation on dry ice for 10 min, the mixture was incubated at 37 °C for 10 min. After three repetitions, the homogenization mixture was centrifuged at 15,000*g* for 15 min. The protein concentration of the supernatant was determined using the Bradford procedure (Bio-Rad, Hercules, CA, USA). The proteins (50 µg whole cell extract) were resolved on 10% SDS–PAGE gels and transferred onto polyvinylidene fluoride (PVDF) membranes. The membranes were incubated with rabbit polyclonal immunoglobulin G (IgG) against OGG1 at a 1:3000 dilution and then were incubated with horseradish peroxidase-labeled antibodies against rabbit IgG (Amersham Biosciences, Piscataway, NJ, USA) at a 1:3000 dilution, and the signals were observed with enhanced chemiluminescence (Amersham Biosciences). To compare the amounts of protein present in the Western assay, blots were reprobed with an anti-β-tubulin antibody (sc-5274, Santa Cruz Biotechnology).

## Results

### *Synthesis and characterization of RF M13mp18-P53 DNA with a unique 8-oxoG at a defined position*

The method of construction of a ccc double-stranded DNA (dsDNA) containing a unique 8-oxoG at a defined site is based on a site-directed mutagenesis procedure using single-stranded DNA (ssDNA) as a template. To achieve higher yields of these types of constructs, we first purified M13mp18-P53 ssDNA and annealed a phosphorylated ODN with a single 8-oxoG (Fig. 1). The 3′ terminus of the ODN acts as primer for complementary strand synthesis by RB69 polymerase. The newly synthesized DNA strand is covalently closed with T4 DNA ligase, producing a circular DNA containing 8-oxoG with a yield of ccc DNA of approximately 50%. CsCl/EtBr gradient purification yields a product with greater than 90% purity of the DNA substrate in the ccc form.

To confirm that the purified ccc M13mp18-P53 DNA contains a single 8-oxoG site, it was incubated with Fpg to introduce a specific nick by the DNA glycosylase–apurinic/apyrimidinic (AP) lyase activity. Following treatment with Fpg, a 5′ primer extension reaction using a $^{32}$P-labeled

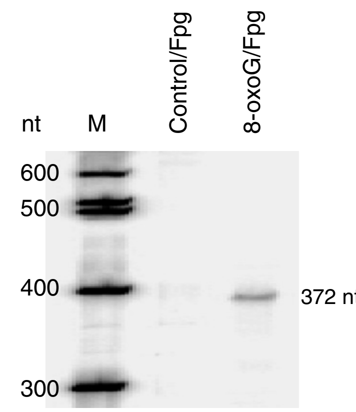


Fig. 2. Primer extension assay for vectors with unique modified sites. Primer extension was carried out using CHM3 (Table 1) for both the undamaged DNA (control/Fpg) and the damaged DNA (8-oxoG/Fpg) when both the control and 8-oxoG-containing RF M13mp18-P53 DNAs have been treated using Fpg. Reaction conditions are described in Materials and methods.

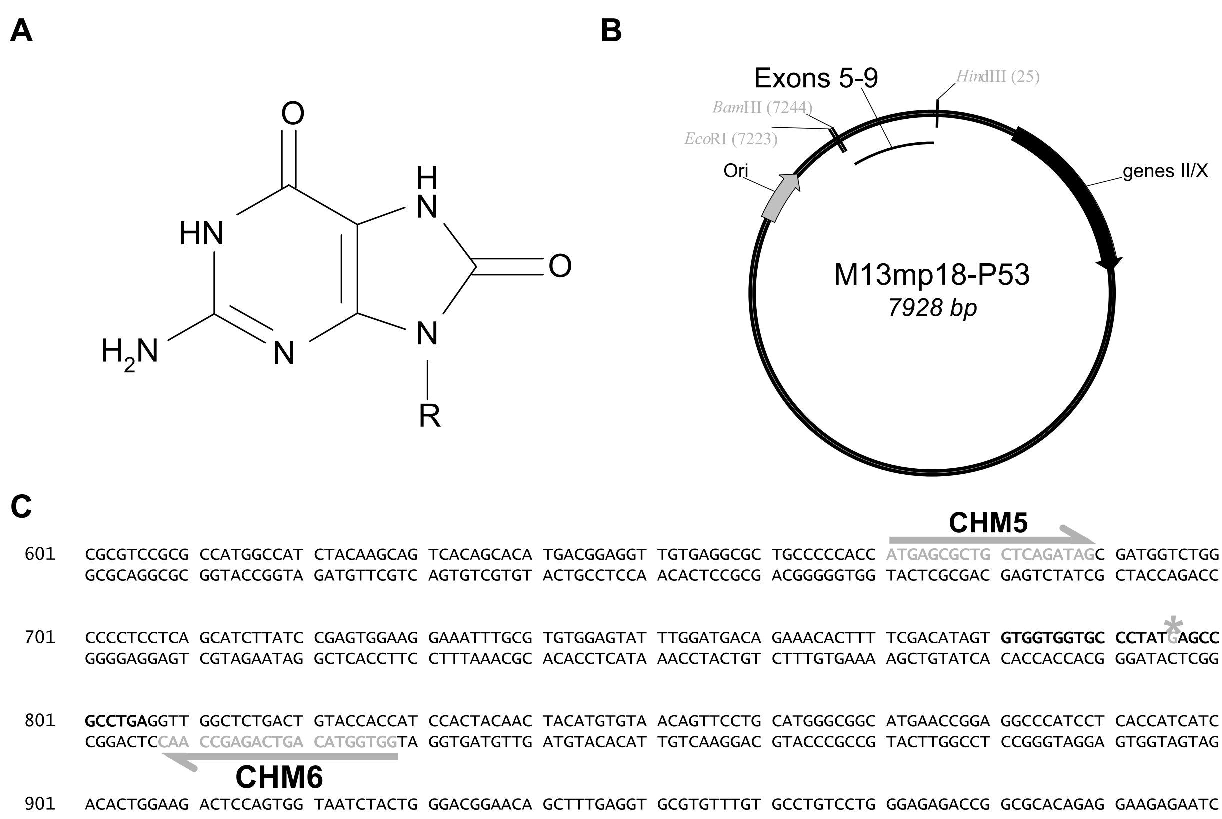


Fig. 1. Damage and vectors used in this study. (A) 8-Oxoguanine. (B) M13mp18-P53 with the cDNA from exons 5 to 9 of human P53. (C) P53 coding sequence from the NCBI database—AF307851 (nucleotide) used in these experiments. The gray arrows and sequences indicate ODNs, the bold black sequence indicates the ODN for 8-oxoG, and the actual position is highlighted in gray with a gray asterisk.

primer demonstrates that the 8-oxoG is in the correct position (Fig. 2). This contrasts with the unmodified DNA in the control reaction that showed only a very weak band following Fpg treatment and primer extension. No damage that blocks primer extension is observed in the region used for amplification in these experiments.

*Assay linearity*

The assay design is outlined in Fig. 3. We hypothesized that damage to recovered ccc DNA would result in a reduction of the available template for PCR (decreased DNA integrity) in the damaged P53 cDNA region. The 8-oxoG is removed from the RF M13mp18-P53-8-oxoG DNA using Fpg. Following the 8-oxoG removal and nick in the ccc DNA, a linear PCR extension is performed using the strand that originally contained the 8-oxoG. The products of the linear extension include repaired DNA that should extend beyond the original position of the 8-oxoG and unrepaired DNA that does not support extension beyond the 8-oxoG position. The use of an aliquot of the primer extension reaction should then yield differences between the repaired and unrepaired DNA. In quantitative real-time PCR, the presence of the 8-oxoG is reflected by

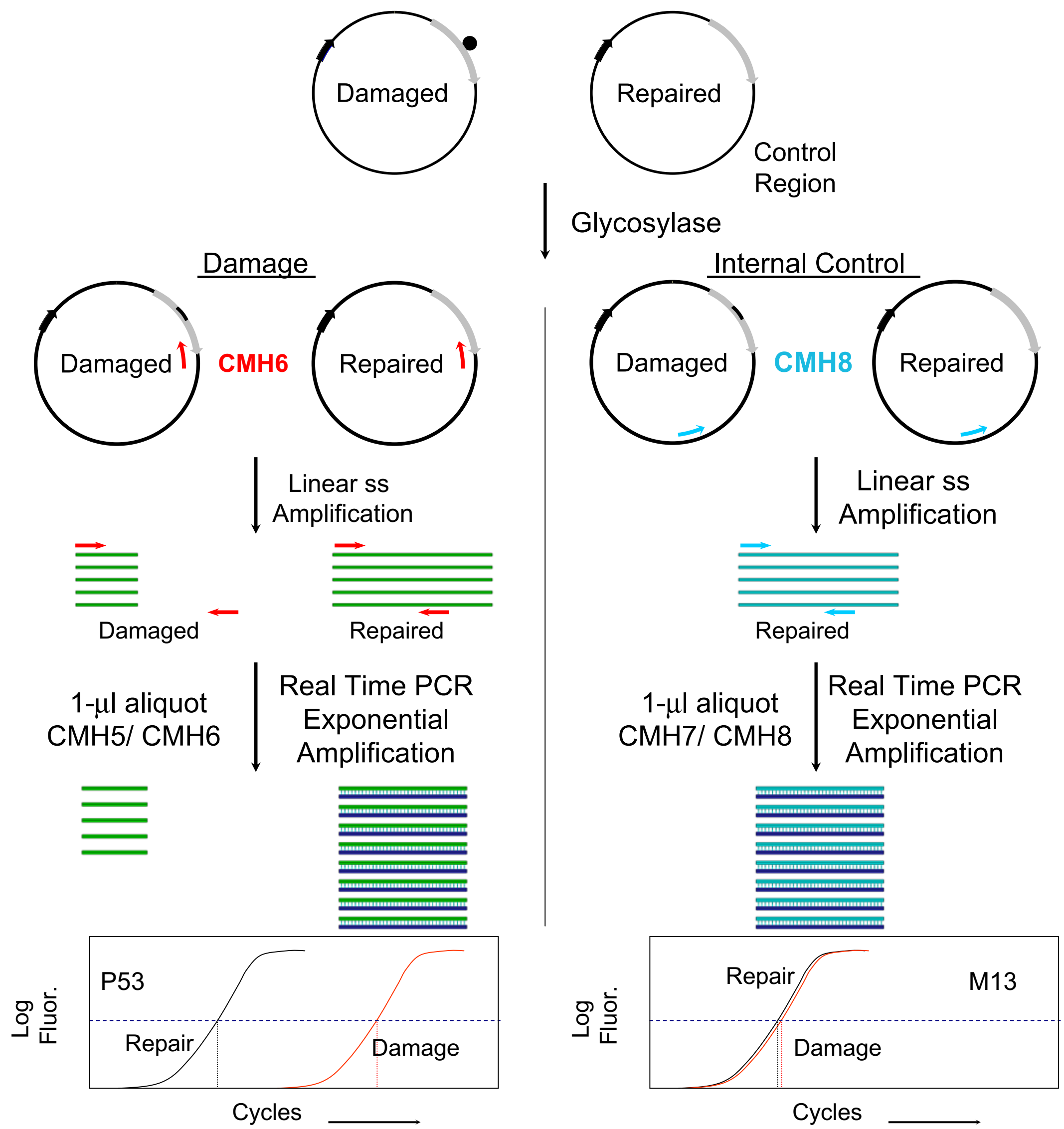


Fig. 3. Outline of method to monitor in vivo repair of 8-oxoG in RF M13mp18-P53. The left side of the figure represents the P53 coding sequence with the 8-oxoG damage, and the right side of the figure represents the control M13 region. Following harvest of episomal DNA, the concentration of DNA obtained is determined using real-time PCR with a nontransfected, undamaged M13 control sequence using a standard curve by amplification of the control region of the M13mp18-P53 (this is not depicted in the figure). The M13mp18-P53 is then reacted with Fpg to remove any remaining 8-oxoG (black circle) and nick the DNA (black arc). A linear PCR extension (15 cycles) with a single ODN is performed using the original damage-containing strand as a template with CHM6 and CHM8 ODN primers. At the end of the extension reaction, a 1-µl aliquot of the product is removed from both reactions, new CHM5 and CHM6 primers are added and new CHM7and CHM8 primers are added to the other tube, and the DNA products of the linear extension reactions (linear amplification) are amplified using exponential real-time PCR in the presence of SYBR Green. ss, single-strand; Fluor., fluorescence.

an increase in the number of cycles to observe an amplification signal. The control region, part of the M13mp18 gene, was unaffected by Fpg treatment. The repair rates of ccc DNA damage were analyzed by calculating the number of cycles for the damaged region (P53 region) divided by the number of cycles for the internal control region (mp13region). The (cycles for undamaged P53 region)/(cycles for M13 region) ratio was defined as 100%, and ccc DNA damage would be expressed as a percentage of this control value. The 0% repair value was set by the (cycles for damaged P53 region)/(cycles for M13 region) ratio. A signal is still obtained from 0% repaired DNA due to the fact that the 1:50 dilution used in the exponential PCR still has the undamaged complementary strand present. Thus, given enough amplification, the fluorescence observed is significant (Figs. 4A and 5A). However, to use this assay for the study of in vivo repair, the assay must be linear. Therefore, we removed the unique 8-oxoG from RF M13mp18-P53-8-oxoG DNA using Fpg and performed a mixing experiment with undamaged RF M13mp18-P53 DNA (Fig. 4 and Table 2). The linear response of triplicate samples in the mixing experiment demonstrates that the assay can be used in repair experiments (Fig. 4 and Table 2).

*Knockdown of OGG1 expression*

The assay method should distinguish between cells that are proficient or deficient in OGG1 expression. OGG1 is a DNA glycosylase that is important in the elimination of 8-oxoG adducts. To test this, we prepared a cell line that was deficient in OGG1 excision of 8-oxoG using siRNA. Fig. 5 shows that there is a significant reduction both in OGG1 messenger RNA (mRNA) levels and in the protein produced for AD293 cells at 2 days posttransfection. The reduction of OGG1 activity is maintained following a second treatment with siRNA 214 at 48 h. Based on these data, the siRNA should decrease the levels in AD293 cells enough to manifest a phenotype with respect to repair of 8-oxoG. One of the three 21-bp oligoribonucleotides, siRNA 214, is the most efficient at the reduction of OGG1 levels (Fig. 5).

*Repair of 8-oxoG in episomal DNA in cells proficient and defective in 8-oxoG repair*

To assess the repair of 8-oxoG in vivo, the RF M13mp18-P53 DNA-8-oxoG was transfected into AD293 cells and cells were harvested from 2 to 48 h posttransfection.

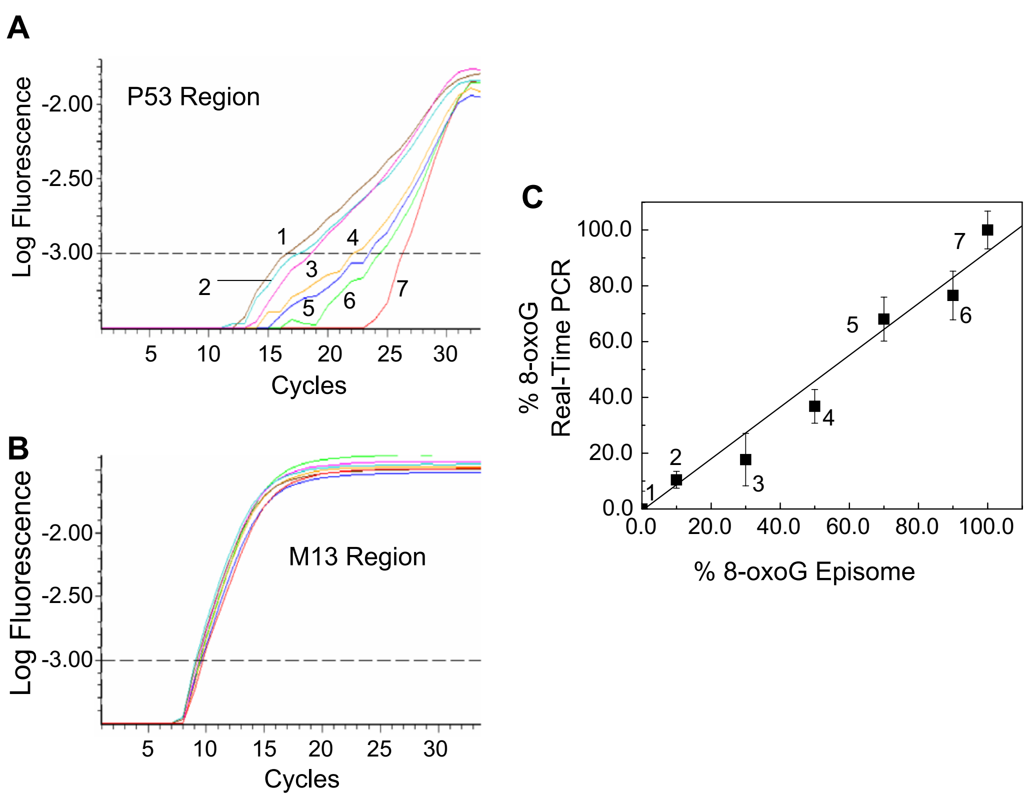


Fig. 4. Linearity of the assay as determined using cleaved and undamaged M13mp18-P53. (A) Log fluorescence versus cycle number obtained by real-time PCR for the repair of 8-oxoG in nicked M13mp18-P53 DNA in AD293 cells in the region of the adduct. The M13mp18-P53 was nicked using Fpg and mixed with M13mp18-P53 without an 8-oxoG. The numbers correspond to the points in panel C. (B) Log fluorescence versus cycle number obtained by real-time PCR of the M13mp18-P53 DNA control region. The colors correspond to the same mix as for the P53 region. The differences in the scans are too close to indicate on the figure. (C) Plot of the mixing curve from 0 to 100% OGG1 (nicked M13mp18-P53). Each point is the mean of triplicate assays, and the standard deviation from the mean for triplicate assays is indicated on the figure. The calculations for the data are presented in Table 2. The correlation coefficient value indicating linearity is 0.987.

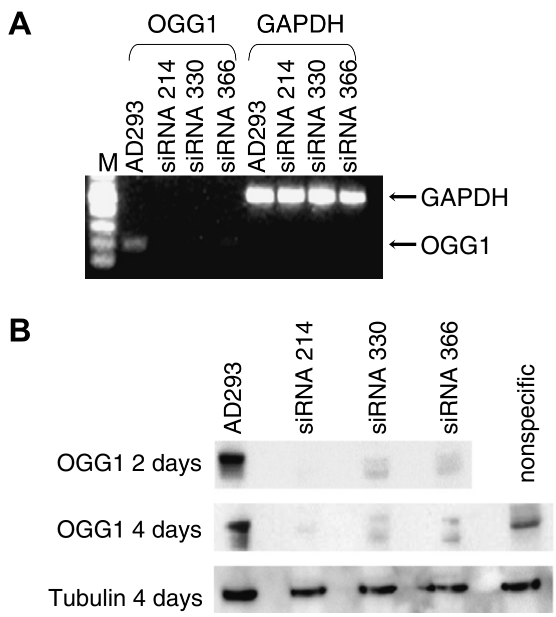


Fig. 5. siRNA knockdown of OGG1 mRNA in AD293 cells. (A) RT–PCR using three different target siRNAs. M corresponds to markers, AD293 corresponds to OGG1 mRNA amplified from those cells, and siRNAs correspond to the different ODNs in the text. The GAPDH amplification is given for the cells as a control. (B) Western blot showing the reduction of OGG1 at 48 and 96 h compared with the production of tubulin. The AD293 cells are without siRNAs, the AD293 cells are transfected with the different siRNAs, and the AD293 cells with the nonspecific siRNAs are also indicated.

From the purified nuclei, the episomal DNA was isolated and subjected to the assay in triplicate. In comparison with this mixing experiment, for repair samples it is necessary to perform the assay on an undamaged template to obtain the value for 100% repair.

Calculations used to monitor 8-oxoG repair are described for the data set for transfection of the RF M13mp18-P53 DNA-8-oxoG into AD193 cells (Table 2). This quantitative PCR assay reveals that in AD293 cells, 8-oxoG is not efficiently removed (only 50% at 24 h after incubation), but at 48 h more than 80% repair is observed (Fig. 6).

We reduced cellular levels of OGG1 for 2 days using the siRNA 214 and then transfected the RF M13mp18-p53 DNA-8-oxoG along with more siRNA 214. Although the AD293 cells repair virtually all of the 8-oxoG in 48 h, there is less than 30% repair observed in the cells that were subjected to siRNA knockdown of OGG1 expression (Fig. 6). This indicates that the assay is sensitive to differences in repair of 8-oxoG in cells. There is some change in the repair levels that could be due to variable OGG1 transcript levels (Fig. 5B, OGG1 4 days), but overall the OGG1 levels remain low during the experiment.

In other experiments, we have performed the assay on AD293 cells that were thawed at a different time, grown, and transfected with the RF M13mp18-P53 DNA-8-oxoG; harvested the ccc DNA; and then performed the assay. Under the same conditions, at least for AD293, there was almost no difference in the observed repair rates. This demonstrates that the technique is robust and repeatable.

*Repair of 8-oxoG normal PBMCs*

To monitor repair in PBMCs, we obtained blood samples from 3 volunteers. The method and strategy are the same as for the AD293 cells. The major difference is that the transfection method uses nucleofection, a modified electroporation technique. Using this method, it is possible to transfect the PBMCs at approximately 40% efficiency. The 3 individuals were transfected with the RF M13mp18-P53-8-oxoG DNA and harvested at different times. Therefore, we can monitor the repair in normal human cells. For 2 individuals, only single blood samples were analyzed, but for a third individual two samples were obtained and show that the half-life for 8-oxoG is similar (Fig. 7). For these individuals, the half-life for 8-oxoG repair varies from 2 to 4 h. Currently, the sampling is too small to yield a range of values for normal individuals, but from these limited experiments 8-oxoG repair is rapid, with half-lives of less than 5 h. One significant point in this article is the capacity to screen for DNA repair defects in individual samples without resorting to stimulation.

## Discussion

In this article, we have detailed a method to monitor DNA repair in mammalian cells and yeast using a simple transfection and quantitative real-time PCR. The method does not require the isolation of large amounts of DNA and is reproducible. Compared with the assays used previously to monitor DNA repair in mammalian cells, this assay requires less manipulation and also provides for triplicate samples [37]. The use of an internal control sequence means that it is not necessary to perform cotransfections or to estimate the transfection efficiency.

The method has a linear response over a range necessary to monitor repair differences in human cells. When used to monitor 8-oxoG repair, the rate was at least twofold slower in AD293 cells than that reported previously in immortalized mouse embryonic fibroblast cells [37] and also slower than repair monitored in the normal human PBMCs in the current study. The AD293 cell line is also deficient in MLH1, so there may be a contribution of mismatch repair to the elimination of 8-oxoG. Previous studies have also shown a role for NER in 8-oxoG elimination [39]. Therefore, the elimination of oxidative damage could be linked to base, nucleotide, and mismatch repair. In the future, this technique could be employed to evaluate the roles of these three repair systems in protection against 8-oxoG.

This technique has some similarity to other techniques that involve PCR-based assays, but all of those techniques introduce a number of sites into either plasmids or genomic DNA. The use of an episome with a unique damage site

Table 2
Calculations from real-time PCR data for mixing of damaged and undamaged RF M13mp18-P53 DNA

| % 8-oxoG | $C_{P53}$[a] | $C_{M13}$[b] | Average $C_{P53}$[c] | $\sigma_{P53}$[d] | Average $C_{M13}$[e] | $\sigma_{M13}$[f] | Average $C_{M13}$ / Average $C_{P53}$[g] | $\sigma_{Ratio}$[h] | Repair (%)[i] | $\sigma_{Repair}$[j] |
|---|---|---|---|---|---|---|---|---|---|---|
| 0.0 | 26.248 | 9.808 | 25.852 | 0.343 | 9.600 | 0.249 | 0.371 | 0.0108 | 0.0 | 6.4 |
| | 25.661 | 9.324 | | | | | | | | |
| | 25.648 | 9.669 | | | | | | | | |
| 10.0 | 24.357 | 9.397 | 24.371 | 0.151 | 9.483 | 0.110 | 0.389 | 0.0051 | 10.5 | 3.0 |
| | 24.529 | 9.443 | | | | | | | | |
| | 24.227 | 9.607 | | | | | | | | |
| 30.0 | 23.463 | 9.536 | 23.198 | 0.781 | 9.311 | 0.197 | 0.401 | 0.0159 | 17.7 | 9.4 |
| | 22.319 | 9.235 | | | | | | | | |
| | 23.811 | 9.164 | | | | | | | | |
| 50.0 | 22.145 | 9.689 | 22.306 | 0.162 | 9.675 | 0.217 | 0.434 | 0.0102 | 36.8 | 6.0 |
| | 22.469 | 9.452 | | | | | | | | |
| | 22.306 | 9.886 | | | | | | | | |
| 70.0 | 19.566 | 9.292 | 19.000 | 0.515 | 9.249 | 0.043 | 0.487 | 0.0133 | 68.1 | 7.9 |
| | 18.562 | 9.250 | | | | | | | | |
| | 18.871 | 9.206 | | | | | | | | |
| 90.0 | 18.327 | 9.082 | 17.934 | 0.361 | 8.988 | 0.194 | 0.501 | 0.0148 | 76.5 | 8.7 |
| | 17.617 | 8.764 | | | | | | | | |
| | 17.858 | 9.118 | | | | | | | | |
| 100.0 | 16.838 | 9.170 | 16.783 | 0.168 | 9.080 | 0.169 | 0.541 | 0.0114 | 100.0 | 6.7 |
| | 16.595 | 9.184 | | | | | | | | |
| | 16.917 | 8.884 | | | | | | | | |

[a] Number of cycles for amplification of the P53 region.
[b] Number of cycles for amplification of the M13 region.
[c] Average number of cycles for amplification of the P53 region.
[d] Standard deviation of cycles for amplification of the P53 region.
[e] Average number of cycles for amplification of the M13 region.
[f] Standard deviation of cycles for amplification of the M13 region.
[g] Ratio of the average cycles for amplification of the M13 region divided by the average cycles for amplification of the P53 region.
[h] Standard deviation for the average cycles for amplification of the M13 region divided by the average cycles for amplification of the P53 region (value in note g multiplied by the square root of $\sigma^2_{CM13}/C^2_{M13} + \sigma^2_{CP53}/C^2_{P53}$, where the Cs are the average values for the amplification of each region).
[i] Repair calculated as 100(average $C_{M13}$ / average $C_{P53}$ – minimum average $C_{M13}$ / $C_{P53}$) / (maximum average $C_{M13}$ / average $C_{P53}$ – minimum average $C_{M13}$ / average $C_{P53}$).
[j] $\sigma_{Repair}$ calculated by placing the standard deviation of the ratios into the equation in note h and taking the difference.

permits the use of other sequences in the episome as an internal standard. This increases the accuracy of the assay and renders it nearly independent of transfection. Previous work using host cell reactivation assays with either luciferase or chloramphenicol acetyltransferase assays has not used this type of internal control sequence. Moreover, this assay directly monitors repair of the adduct in DNA, unlike the host cell reactivation assays that depend on the synthesis of a functional protein.

Because this method depends on the amplification of a single strand of DNA, we should be able to adapt this method to study transcription effects. DNA repair linked to transcription is difficult to examine using DNA randomly damaged at multiple sites in episomes. With this assay, it is possible to investigate damage on either transcribed or nontranscribed strands. This can be critical in some experiments; for example, repair of 8-oxoG by OGG1 has been linked to repair primarily on the nontranscribed strand, and that would not be observable using conditions for most host cell reactivation assays [37].

Despite the advantage of the assay, the preparation of the substrate is more difficult than that of host cell reactivation assays. The requirement to have a unique damaged site means that preparation of the episome is more challenging than in most host cell reactivation experiments that are based on the observation of protein activity by damaging an episome globally. The synthesis of large amounts of modified DNAs is necessary, but the use of both the DNA polymerase and DNA ligase synthesized in the laboratory facilitates this task. To obtain the quantities of modified DNA, we often perform a number of reactions in parallel to ensure that there will be enough ccc DNA to use. Even with this hurdle, the fact that the episomal DNA has its own internal standard in this assay provides a distinct advantage and yields data on the repair capacity in cells that do not depend on preparing cell extracts [22,55].

Host cell reactivation assays have also been used to investigate PBMCs. Those are sensitive but lack the specificity to examine the repair kinetics of the adducts

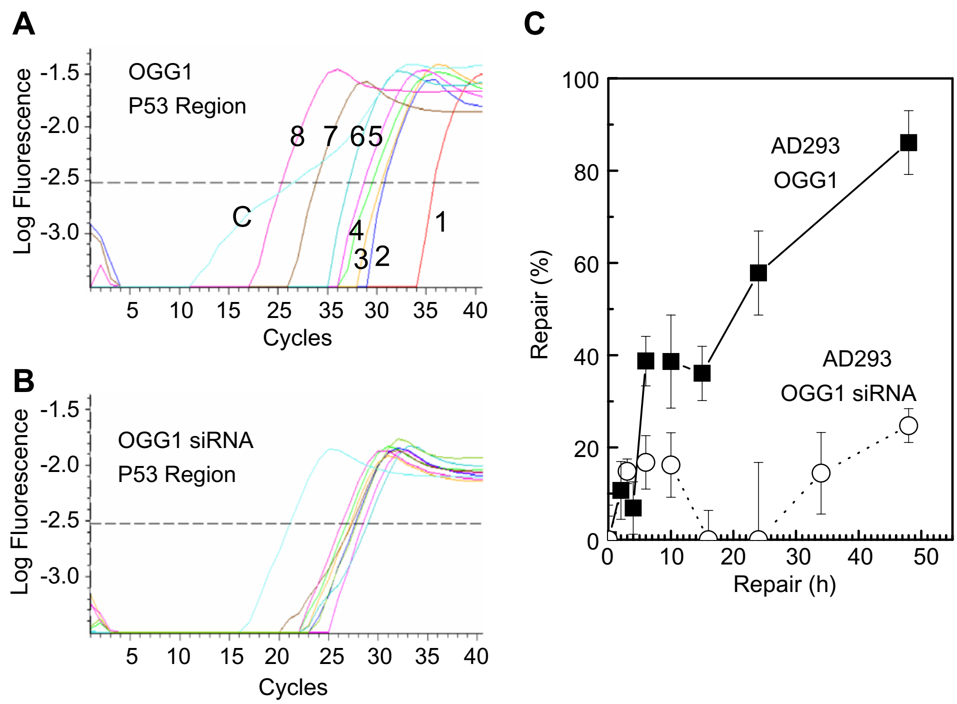


Fig. 6. Repair of 8-oxoG following transfection into a human cell line. The concentration of ccc DNA was determined using real-time PCR comparison with a standard curve. Harvested DNA was treated with Fpg to introduce a nick in all unrepaired DNA. (A) Real-time PCR results for the repair of 8-oxoG in M13mp18-P53 DNA in AD293 cells in the region of the adduct. The numbers correspond to the following times: 1, 0 h (100% damaged); 2, 2 h; 3, 4 h; 4, 6 h; 5, 10 h; 6, 15 h; 7, 24 h; 8, 48 h. The curve is fitted to a sigmoidal function. C represents the control undamaged DNA that corresponds to 100% repair. The dotted line represents the points in each of the curves at which the cycle number was obtained. (B) Real-time PCR results for repair of the M13mp18-P53 DNA in AD293 cells with the OGG1 siRNA knockdown. The dotted line represents the points in each of the curves at which the cycle number was obtained. (C) Repair determined using real-time PCR. Plots of the percentage repair as a function of time are indicated. Repair of 8-oxoG in AD293 cells is represented by closed squares, and repair in AD293 cells with siRNA knockdown of OGG1 is represented by open circles. Each point is the mean of triplicate assays, and the standard deviation from the mean for triplicate assays is indicated on the figure.

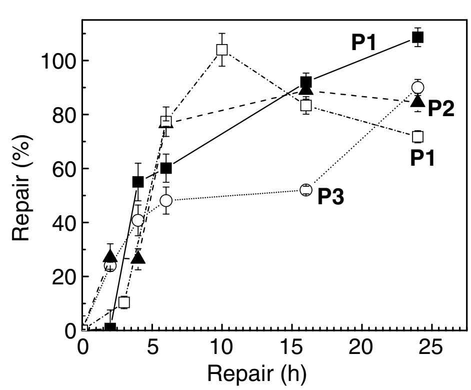


Fig. 7. Repair of 8-oxoG following transfection into human white blood cells. ccc DNA harvested as a function of time following transfection into human PBMCs is treated as in Fig. 6. The numbers represent individual samples. Each transfection used $1 \times 10^6$ PBMCs. Plots represent the percentage repair as a function of time for each of the different patient samples. Each point is the mean of triplicate assays, and the standard deviation from the mean for triplicate assays is indicated on the figure. P1, P2, and P3 refer to the individuals assayed. P1 was assayed twice on different days.

induced because a property of a synthesized protein usually is detected. The sensitivity of this assay, the possibility of obtaining data easily, and the assay's reproducibility coupled with the use of an efficient means to transfect lymphocytes make this an ideal system for the investigation of DNA repair in patient samples. With several modifications, it should also be adaptable to multiplexing. This method should prove to be useful in studies of populations that have potential DNA repair defects and should be adaptable to many types of DNA damage.

## Acknowledgments

The authors thank William Konigsberg for the RB69 overproducing clone, Xiangrong Zhao for preparation of the RB69 DNA polymerase protein, and Stephen Bates for assistance with the cell culture. This investigation was funded by the City of Hope, a SPORE (Specialized Program of Research Excellence) grant (Stephen Forman), and the National Cancer Institute (T.R.O. and R.B.).

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